Getting Started with PostgreSQL Shell

Quick Command Reference

\list	Show all databases
\du	Show all user accounts
\c <database>	Switch to using the specified database
\dt	List tables in database
\d <table>	See details of table

Download and Install PostgreSQL

https://www.postgresql.org/download/

Introductory Exercise

Create a new user and give the user permissions

Log in as user postgres, using the password you created during installation.

Create a database.

```
CREATE DATABASE pets;
```

Create a user and this user complete access to the pet database. Be sure to use this username and password for this course because the TAs will create a user with the same name and password. This will allow them to use your code on their system without needing to change credentials.

```
CREATE USER oop password 'ucalgary';
ALTER DATABASE pets OWNER TO oop;
GRANT ALL PRIVILEGES ON DATABASE pets TO oop;
```

Explore the database as the new user

Log in as user oop, using the password ucalgary.

Switch to using the pet database.

```
\c pets
```

Create a table in the database.

```
CREATE TABLE cat
(
  id        SERIAL PRIMARY KEY,
  name      VARCHAR(150) NOT NULL,
  owner     VARCHAR(150) NOT NULL,
  birth     DATE NOT NULL,
  death     DATE
);
```

View the tables in the database.

```
\dt
```

View the details of the cat table.

```
\d cat
```

Insert some data in the table.

```
INSERT INTO cat ( name, owner, birth) VALUES
  ( 'Sandy', 'Jane', '2020-01-03' ),
  ( 'Cookie Destroyer of Worlds', 'Chandrakala', '2023-11-13' ),
  ( 'Sleepy', 'Sooyoon', '2019-05-21' );
```

View a subset of the data.

```
SELECT name FROM cat WHERE owner = 'Jane';
```

Delete some data.

```
DELETE FROM cat WHERE name='Sleepy';
```

View the data in the table.

```
SELECT * FROM cat;
```

Disconnect from postgres.

```
exit
```

Note

If you do not log out of the PostgreSQL terminal properly, you may subsequently find that you cannot perform some operations because users are still accessing it. When this happens, you will need to restart PostgreSQL. This command will vary by system. On my system (macOS Sonoma 14.2.1 with PostgreSQL 16), the command is:

```
sudo -u postgres /Library/PostgreSQL/16/bin/pg_ctl -D
/Library/PostgreSQL/16/data restart
```

But it is better to avoid this situation by properly disconnecting.

Clean up

Now that we have finished this demonstration, let's clean up by removing the database as user postgres. We can also just remove permissions.

```
ALTER DATABASE pets OWNER TO postgres;
REVOKE ALL PRIVILEGES ON DATABASE pets FROM oop;
DROP DATABASE pet;
```

Log out from the postgres shell.

```
exit
```

Tips

- Only users with admin privileges can create databases. To give another user access to a database, you must create the database then grant access to the other user (as admin).
- When you create tables as an admin, other users will not have access to the table, even if the other users have access to the database. You will need to explicitly grant access to each table. If you plan on working with the database as another user, you may find it easier to log in as the other user and create the tables as the other user.
- If tables have dependencies and cannot be deleted, you can add `CASCADE` at the end of the delete statement to allow for deletion despite constraints.